

Worksheets W1–W4

accompanying *Mild p -class tower groups of imaginary quadratic fields*

These worksheets contain the worked examples referenced from the paper as Parts W1–W4; the tables of Parts W5 and W6 are separate documents in this directory. References of the form Proposition 2.12, §4.2, Section 5, or (P. n) refer to statements, sections, and numbered equations of the paper; plain parenthesised numbers refer to equations of this document.

Part W1: The principal example — class fields and secondary norm operators

This part and Part W2 record the complete data of the first field of Section 5 of the paper (the complete $p = 5$ table is Part W5). We specialize to

$$K = \mathbb{Q}(\sqrt{-2800905}), \quad p = 5. \quad (1)$$

The field has discriminant

$$\text{disc}(K) = -11203620$$

and

$$\text{Cl}(K) \simeq (\mathbb{Z}/10\mathbb{Z})^3. \quad (2)$$

Fix generators g_1, g_2, g_3 for the three cyclic factors in (2). Their images $\overline{g_1}, \overline{g_2}, \overline{g_3}$ in $\text{Cl}(K)/5$ form a basis, and we let

$$\chi_1, \chi_2, \chi_3 \in (\text{Cl}(K)/5)^* \simeq H^1(G, \mathbb{F}_5)$$

be the dual character basis, so that for example $\chi_1(\overline{g_1}) = 1$ and $\chi_1(\overline{g_2}) = \chi_1(\overline{g_3}) = 0$. On the torsion side we use

$$e_1 = 2g_1, \quad e_2 = 2g_2, \quad e_3 = 2g_3, \quad (3)$$

which is a basis of $\text{Cl}(K)[5]$. These are separate choices of bases of $\text{Cl}(K)/5$ and $\text{Cl}(K)[5]$; no identification between these two $\mathbb{F}_5$ -vector spaces is being made.

The reconstruction theorem requires the secondary norm operators attached to the three basis characters and their three pairwise sums, namely

$$\chi_1, \chi_2, \chi_3, \chi_1 + \chi_2, \chi_1 + \chi_3, \chi_2 + \chi_3. \quad (4)$$

We first record explicitly the six corresponding unramified cyclic extensions. For a character x , the vector $\boldsymbol{\eta}$ in the second column is the integer character used by PARI on $\text{Cl}(K) \simeq (\mathbb{Z}/10\mathbb{Z})^3$. Since each cyclic factor has order 10, the PARI vector is obtained by multiplying the coordinates of x by $10/5 = 2$. Its kernel is an index-5 subgroup of $\text{Cl}(K)$ whose fixed field is L_x .

The last column lists generators of this subgroup directly, in the fixed coordinates determined by g_1, g_2, g_3 . Thus, for example, the first row says that the kernel belonging to χ_1 is generated by $(5, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ in $(\mathbb{Z}/10\mathbb{Z})^3$.

x	PARI vector $\boldsymbol{\eta}$	generators of the class-field kernel
χ_1	$[2, 0, 0]$	$(5, 0, 0), (0, 1, 0), (0, 0, 1)$
χ_2	$[0, 2, 0]$	$(1, 0, 0), (0, 5, 0), (0, 0, 1)$
χ_3	$[0, 0, 2]$	$(1, 0, 0), (0, 1, 0), (0, 0, 5)$
$\chi_1 + \chi_2$	$[2, 2, 0]$	$(5, 0, 0), (4, 1, 0), (0, 0, 1)$
$\chi_1 + \chi_3$	$[2, 0, 2]$	$(5, 0, 0), (0, 1, 0), (4, 0, 1)$
$\chi_2 + \chi_3$	$[0, 2, 2]$	$(1, 0, 0), (0, 5, 0), (0, 4, 1)$

Each displayed subgroup has index 5, so its class field has degree 5 over K . In all six cases the finite relative discriminant is trivial. Hence the extensions $L_x|K$ are unramified at every finite prime; since K is imaginary quadratic, they are everywhere unramified.

For the Ahlqvist–Carlson formula we need more than the field L_x : the generator $\sigma_x \in \text{Gal}(L_x|K)$ must be normalized by the actual character x . We therefore also record the normalization data used in the computation, in the notation of §4.2 of the paper: σ_{Artin} denotes the cyclic generator of $\text{Gal}(L_x|K)$ chosen by the class-field computation, and the vector $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ is defined by

$$\text{Artin}(g_i) = \sigma_{\text{Artin}}^{\alpha_i}.$$

The scalar λ is characterized by $\alpha_i = \lambda x(\overline{g_i})$ in $\mathbb{F}_5$; the scalar u compares the generator initially used by the implementation with σ_{Artin} , and ν_x is defined by $\sigma_x = \sigma_{\text{Artin}}^{\nu_x}$. Thus ν_x is the exponent of the finally normalized generator. In the present computation $u = 1$ in every case, so $\nu_x = \lambda$.

x	α	λ	u	ν_x
χ_1	$[3, 0, 0]$	3	1	3
χ_2	$[0, 4, 0]$	4	1	4
χ_3	$[0, 0, 4]$	4	1	4
$\chi_1 + \chi_2$	$[3, 3, 0]$	3	1	3
$\chi_1 + \chi_3$	$[3, 0, 3]$	3	1	3
$\chi_2 + \chi_3$	$[0, 2, 2]$	2	1	2

As a direct check of the normalization, for every row and every $i = 1, 2, 3$ we verified

$$\text{Artin}(g_i) = \sigma_x^{x(\overline{g_i})}.$$

Proposition 1 (Secondary norm matrices). *Represent an element of $\text{Cl}(K)/5$ by its coordinate column vector with respect to $\overline{g_1}, \overline{g_2}, \overline{g_3}$. With respect to the input basis e_1, e_2, e_3 of $\text{Cl}(K)[5]$, the six secondary norm operators $D_x: \text{Cl}(K)[5] \rightarrow \text{Cl}(K)/5$ are represented by the matrices*

$$\begin{aligned}
D_{\chi_1} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 3 & 0 \end{pmatrix}, & D_{\chi_2} &= \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \\ 4 & 2 & 4 \end{pmatrix}, & D_{\chi_3} &= \begin{pmatrix} 1 & 2 & 0 \\ 4 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \\
D_{\chi_1 + \chi_2} &= \begin{pmatrix} 0 & 3 & 1 \\ 0 & 2 & 4 \\ 0 & 1 & 2 \end{pmatrix}, & D_{\chi_1 + \chi_3} &= \begin{pmatrix} 1 & 4 & 0 \\ 1 & 4 & 4 \\ 4 & 1 & 0 \end{pmatrix}, & D_{\chi_2 + \chi_3} &= \begin{pmatrix} 3 & 1 & 2 \\ 0 & 3 & 2 \\ 0 & 2 & 3 \end{pmatrix}.
\end{aligned}$$

Proof. For each of the 18 pairs (x, e_j) , the constructed pair $(t_{x,j}, I'_{x,j})$ satisfies the two exact equations (P.46)–(P.47). By Proposition 2.12 of the paper, these verified equations imply

$$D_x(e_j) = [N_x(I'_{x,j})] \in \text{Cl}(K)/5.$$

The norm ideal on the right is recomputed independently and its class is expressed in the coordinates $\overline{g_1}, \overline{g_2}, \overline{g_3}$. The resulting values are

x	$D_x(e_1)$	$D_x(e_2)$	$D_x(e_3)$
χ_1	$[0, 0, 0]$	$[0, 3, 3]$	$[0, 1, 0]$
χ_2	$[0, 0, 4]$	$[1, 0, 2]$	$[2, 0, 4]$
χ_3	$[1, 4, 0]$	$[2, 0, 0]$	$[0, 1, 0]$
$\chi_1 + \chi_2$	$[0, 0, 0]$	$[3, 2, 1]$	$[1, 4, 2]$
$\chi_1 + \chi_3$	$[1, 1, 4]$	$[4, 4, 1]$	$[0, 4, 0]$
$\chi_2 + \chi_3$	$[3, 0, 0]$	$[1, 3, 2]$	$[2, 2, 3]$

For example, the second entry in the row for χ_1 says

$$D_{\chi_1}(e_2) = 3\overline{g_2} + 3\overline{g_3} \in \text{Cl}(K)/5,$$

and this is the second column of the displayed matrix D_{χ_1} . Reading the three values in each row as coordinate columns gives the six matrices above. The matrices were assembled only after all norm classes had been computed and independently extracted. $\square$

As an additional arithmetic check, we repeated the complete class-field and secondary-norm computation for the three doubled characters and obtained

$$D_{2\chi_1} = 4D_{\chi_1}, \quad D_{2\chi_2} = 4D_{\chi_2}, \quad D_{2\chi_3} = 4D_{\chi_3}. \quad (5)$$

By the quadratic dependence (P.14) these identities are forced; confirming them establishes nothing new, and their worth lies in the fact that they could have failed. The pairs x and $2x$ define the same class-field kernel but require different normalized automorphisms, and a computation that attached the automorphism to $\ker x$ rather than to x would return $D_{2x} = D_x$ here. The check therefore excludes one specific way for the implementation to go wrong at its most delicate step, and no more. What it probes belongs to the procedure rather than to the field, so we applied this check only to the principal example.

Part W2: The principal example — the transversality computation

We apply Corollary 3.15 of the paper to

$$x = a + b.$$

From the fourth matrix in Proposition 1,

$$\text{rk } D_{\chi_1+\chi_2} = 1,$$

with

$$\ker D_{\chi_1+\chi_2} = \langle e_1, 3e_2 + e_3 \rangle$$

and

$$\text{im } D_{\chi_1+\chi_2} = \langle \overline{g_1} + 4\overline{g_2} + 2\overline{g_3} \rangle.$$

Take $v_1 = a$, $v_2 = c$, and $y = b + 3c$. The character y annihilates the image line above, while its restriction to $(a + b)^\perp$ is nonzero. With the adapted datum

$$\mathcal{B} = (e_1, 3e_2 + e_3; a, c; b + 3c),$$

the transversality matrix is

$$B_{a+b, \mathcal{B}} = \begin{pmatrix} 3 & 2 \\ 3 & 4 \end{pmatrix}, \quad \det B_{a+b, \mathcal{B}} = 1 \in \mathbb{F}_5.$$

Thus $\chi_1 + \chi_2$ is transverse, and Theorem 3.13 of the paper proves that $G_\emptyset(K)(5)$ is mild, and hence has cohomological dimension 2.

As an independent check, the six matrices also reconstruct a cubic word matrix of rank three. A direct change of variables and relation basis gives three relations with combinatorially free leading words

$$\text{ZZX}, \quad \text{ZYY}, \quad \text{ZYG}.$$

The accompanying computation checks the shuffle identities, the word order, and Anick's non-overlap condition directly. These additional calculations are not needed for the preceding proof.

Part W3: The norm-degeneracy scheme of the principal example

The norm-degeneracy scheme of Definition 3.17 of the paper can be written out completely for the principal example. With $x = ua + vb + wc$ in the basis of Part W1, polarizing the six matrices of Proposition 1 gives the matrix of quadratic forms

$$D_{(u,v,w)} = \begin{pmatrix} 2vw + w^2 & 2uv + 2uw + v^2 + 3vw + 2w^2 & 4uv + 2v^2 \\ 2uw + vw + 4w^2 & 3u^2 + 4uv + uw + 3vw & u^2 + 3uv + 2uw + vw + w^2 \\ uv + 4uw + 4v^2 + vw & 3u^2 + uv + 3uw + 2v^2 & 3uv + 4v^2 + 4vw \end{pmatrix},$$

whose j th column is $D_x(e_j)$ in the coordinates $\overline{g_1}, \overline{g_2}, \overline{g_3}$. The scheme $\Sigma_D \subseteq \mathbb{P}_{\mathbb{F}_5}^2$ is defined by the nine 2×2 minors, the homogeneous quartics

$$\begin{aligned} & 2u^2vw + 4u^2w^2 + 4uv^2w + 4uw^3 + 4v^3w + 4v^2w^2 + 4vw^3 + 2w^4, \\ & 4u^2vw + u^2w^2 + 3uv^2w + uvw^2 + 2uw^3 + 3v^3w + 4v^2w^2 + 3vw^3 + w^4, \\ & 2u^3w + 4u^2vw + u^2w^2 + 4uv^2w + uvw^2 + uw^3 + 4v^2w^2 + 2w^4, \\ & 3u^2v^2 + u^2vw + uv^3 + uvw^2 + v^4 + v^3w + v^2w^2 + 3vw^3, \\ & u^2v^2 + 4u^2vw + 2uv^3 + 4uv^2w + 3uvw^2 + 2v^4 + v^3w + 2v^2w^2 + 4vw^3, \\ & 3u^3v + u^2v^2 + 4u^2vw + uv^3 + 4uv^2w + 4uvw^2 + v^3w + 3vw^3, \\ & 2u^3v + 4u^3w + 4u^2v^2 + 4u^2w^2 + 4uv^3 + 4uv^2w + 4uvw^2 + 2uw^3, \\ & 4u^3v + u^3w + 3u^2v^2 + u^2vw + 2u^2w^2 + 3uv^3 + 4uv^2w + 3uvw^2 + uw^3, \\ & 2u^4 + 4u^3v + u^3w + 4u^2v^2 + u^2vw + u^2w^2 + 4uv^2w + 2uw^3. \end{aligned}$$

Its closed points can be listed explicitly. On the chart $w = 1$ the radical of the minor ideal has a reduced coordinate ring of $\mathbb{F}_5$ -dimension three, which a single closed point of degree three exhausts; on the line $w = 0$ the ideal restricts to a squarefree quadratic with two rational roots, and $(1:0:0)$ does not lie on Σ_D . Thus Σ_D has exactly three closed points: the rational points $(1:1:0)$ and $(1:3:0)$ and one point of degree three. At $(1:1:0)$ — the character $\chi_1 + \chi_2$ of Part W2 — the tangent space is zero: the point is reduced and isolated, which is the geometric form (Corollary 3.23 of the paper) of the determinant computation carried out there. At $(1:3:0)$ the operator also has rank one, but the tangent space is one-dimensional, so this point is not transverse: rank one alone does not suffice for mildness. The degree-three point is again reduced and isolated, with transversality over its residue field $\mathbb{F}_{125}$; either of the two transverse points proves Theorem 5.2 of the paper for this field. The display is generated deterministically from the stored matrices by `tools/sigma_d_equations.py`.

Part W4: One certificate entry, item by item

To make the certificate format concrete, we follow a single entry from prescribed character to certified value. The table abbreviates two bulky items: the ten elements of B_L are described but not listed, and the 128 factors of t are counted but not printed. The autogenerated companion `certificate-guide.pdf`, included alongside this document, restates all 18 entries of this field in the notation of the paper — it, too, keeps t in compact form —, and the data file `certificate.gp` is the certificate itself, byte for byte. Entry 2 certifies the value $D_{\chi_1}(e_2)$, that is, the second column of the first matrix in Proposition 1.

The rows follow the entry tuple of the certificate definition in the paper's appendix C; the table shows the stored data, lightly annotated (norms, factor counts, residue degrees), and the checks the verifier runs on them are listed after it.

tuple item	stored data (entry 2)
x	the character vector $(1, 0, 0)$, that is, $x = \chi_1$; stored under the label a
e	the basis element e_2 , stored as the column index 2
f_{rel}	$y^5 - y^4 - 1027y^3 + (8s - 33869)y^2 + (340s - 648684)y + (2688s - 3346560)$ over $K = \mathbb{Q}(s)$, $s^2 + 2800905 = 0$
f_{abs}	$y^{10} - 2y^9 - 2053y^8 - 65684y^7 - 174901y^6 + 64171174y^5 + 2665457137y^4 + 66051314232y^3 + 1091726153376y^2 + 9461326049280y + 31436965969920$ over $\mathbb{Q}$
B_L	ten elements of $\mathbb{Q}[\theta]$, θ the image of y in $\mathbb{Q}[y]/(f_{\text{abs}})$, each written as a rational number or a polynomial in θ ; the basis of K is stored once in the certificate header
w	the coordinate vector $(0, -1, 0, 0, 0, 0, 0, -1, 0)$ with respect to B_L
a', J	$a' = \frac{41298191}{2498705294406001} + \frac{16828}{2498705294406001}s$ and $J = \begin{pmatrix} 1201 & 1072 \\ 0 & 1 \end{pmatrix}$ in Hermite normal form, $N(J) = 1201$
t, I'	t in compact form, a product of 128 factors with signed exponents, never expanded; I' an ideal of L_x of absolute norm $10587476640 = 2^5 \cdot 3 \cdot 5 \cdot 13 \cdot 1696711$, in Hermite normal form
$\ell, \mathfrak{q}$	$\ell = 11$ and a degree-one prime $\mathfrak{q}$ above it
v	$(0, 3, 3)$

On these data the verifier decides:

- the change of basis between B_L and a recomputed integral basis of the absolute model is integral with determinant ± 1 — an integral basis is not canonical, and this check is what lets the stored coordinates denote the same algebraic objects everywhere;
- f_{abs} is the absolute defining polynomial derived from f_{rel} , the extension $L_x \mid K$ is of degree 5 and unramified everywhere, and the exact Artin kernel check identifies L_x as the fixed field of $\ker x$;
- the element with coordinate vector w is the image of the primitive element under an automorphism of exact order 5 fixing K , normalized to x as in (P.22) through exact Artin exponents;
- $(a')J^5 = \mathcal{O}_K$ — the condition in (P.4) written multiplicatively — and $[J]$ represents e_2 ;
- $I'^{(1-\sigma_x)^2}(t)J\mathcal{O}_{L_x} = \mathcal{O}_{L_x}$, identity (P.46), holds as an equality of fractional ideals, evaluated factor by factor from the compact t ;
- the principal ideal generated by $N_x(t)/a'$ equals $\mathcal{O}_K$ — computed, not inferred from (P.46) — and the sign is resolved as in Lemma 4.2 of the paper: since the torsion units of K are ± 1 , the reduction at $\mathfrak{q}$, which equals the residue of $+1$, proves $N_x(t) = a'$, identity (P.47);
- the recomputed class $[N_x(I')]$ has coordinate vector $v = (0, 3, 3)$ in $\text{Cl}(K)/5$; by (P.26) this is $D_{\chi_1}(e_2)$, the second column of D_{χ_1} in Proposition 1.

Entries 1–18 certify the six matrices of Proposition 1 column by column in this way.

Parts W5 and W6: the tables

The complete tables — Part W5 for $p = 5$ and Part W6 for $p = 3$ — are the separate documents `tables-p5.pdf` and `tables-p3.pdf` in this directory. Part W6 is generated from the result records of the repository by `generate_tables_p3.py`; regenerating it after an extension of the range reproduces the tables from the records alone.